

L1.7b Related Rates

Work in your group. Every problem gets the full template.

Rates linked by an equation

1. Take $\frac{d}{dt}$ of both sides. Every letter is a function of t . **Stop there — there is nothing to solve.**

(a) $x^2 + y^2 = 100$

(b) $V = \frac{4}{3}\pi r^3$

(c) $y^2x = 4$

(d) $A = \frac{1}{2}bh$

The related-rates template

2. A snowball melts so that its surface area decreases at $1 \text{ cm}^2/\text{min}$. Find the rate at which the diameter decreases when the diameter is 10 cm.

Fill in only the Define box and the ? box. Do not solve anything.

Define

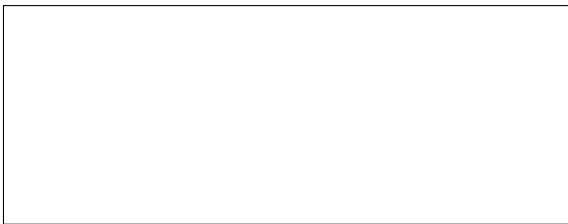
C	_____
V	_____
R	_____

?

Stop. One of these numbers is not a constant. Which one, and why?

3. A circle is shrinking so that its radius contracts at 1 cm/s. How fast is its area changing when $r = 10$ cm?

Picture



Define

C	_____
V	_____
R	_____

?

Solve:

MEQ:

REQ:

4. A kite 100 ft above the ground moves horizontally at 8 ft/s. At what rate is the angle between the string and the horizontal decreasing when 200 ft of string has been let out?

Picture



Define

C	_____
V	_____
R	_____

?

Solve:

MEQ:

REQ:

Worked related-rates problems

5. A Ferris wheel of radius 10 m turns once every 2 minutes, and the bottom of the wheel is at ground level. How fast is a rider rising when the seat is 16 m above the ground?

Picture



Define

C	_____
V	_____
R	_____

?

Solve:

MEQ:

REQ:

Auxiliary equations

6. Gravel is dumped from a conveyor belt at $30 \text{ ft}^3/\text{min}$. It forms a cone whose base diameter and height are **always equal**. How fast is the height increasing when the pile is 10 ft high?

The volume of a cone is $V = \frac{1}{3}\pi r^2 h$.

Picture



Define

C	_____
V	_____
R	_____

?

Solve:

MEQ:

REQ:

Stop. Count the variables in your MEQ whose rate you do not know. If there is more than one, what do you need before you differentiate?

7. **BOARDS** Your friends do not read the rest of the example and try this one on their own:

A ladder 10 ft long rests against a vertical wall. If the bottom of the ladder slides away from the wall at 1 ft/s, how fast is the top of the ladder sliding down the wall when the bottom of the ladder is 6 ft from the wall?

Here is their work, exactly as they wrote it:

“We want to find $\dot{y}$. So we set up

$$x^2 + y^2 = 100$$

Now we want $\dot{y}$ when $\dot{x} = 1$ and $x = 6$. Substituting $x = 6$ gives us

$$36 + y^2 = 100 \quad \text{or} \quad y^2 = 64$$

Now we take derivatives:

$$2y \dot{y} = 0$$

giving $\dot{y} = 0$.”

- (a) **Why does their answer not make sense?**
- (b) **What error did they make, and where exactly does it happen?**
- (c) **Fix it. Which box of the template did they never fill in?**